

Aurora-Lens: Proof & Precedence

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This dossier is a dated public chronology and evidence record for the Aurora / PEF admissibility architecture and its implementation, Aurora-Lens. Its purpose is narrow: to establish chronology, authorship, and public traceability through artefacts that can be dated, checked, and independently inspected. Commercial uptake, institutional approval, and market visibility are separate questions and are not the measure of conceptual origin.

The product claim is runtime governance. The precedence claim is that the architecture, terminology, implementation surface, and public record existed before later market convergence.

What is claimed — and not claimed

CLAIMED	Public authorship and dated precedence of a specific architecture centred on persistent existence, admissibility before consequence, governed refusal, lawful continuation, and auditability.
NOT CLAIMED	This dossier does not prove direct derivation by any later vendor feature. That would require a denser evidentiary chain.
DISTINCTION	Adjacent market convergence does not equal architectural equivalence; nor does it erase prior public authorship.
DISSEMINATION	The presence of a recipient in a dissemination artefact means only that material was sent. It is not represented as endorsement, validation, review, collaboration, or affiliation.

Chronology of externally anchorable milestones

Externally anchorable milestones only; the underlying intellectual development predates this sequence.

20 Nov 2025	Conceptual Bridges Archive repository (Artefact 1).
27 Nov 2025	Provisional patent application 2025905835 filed: AURORA - Autonomous Conceptual Reasoning Architecture.
28 Nov 2025	Provisional patent application 2025905860 filed: PEF Specification.
29 Nov 2025	Provisional patent application 2025905885 filed: Autonomous Compositional Reasoning Architecture Integrated with Persistent Existence Frame.
30 Nov 2025	aurora-demo repository created through GitHub's web interface; implementation files, LEGAL_NOTICE.md, and v1.0.0 tag recorded in GitHub-signed commits (Artefacts 3, 4, 7, 8).
1 Dec 2025	Empirical demonstration document records comparative ambiguity testing tied to the filed provisional applications.
4 Dec 2025	Aurora-PEF repository created. Root commit 0508425 creates the README naming Aurora + PEF and the Persistent Existence Frame (Artefact 2).
6 Dec 2025	aurora-demo made public; direct circulation to researchers begins (Artefact 5).
6-19 Dec 2025	Public exposure window: 556 clone events from 324 unique sources (Artefacts 6, 9).
8 Dec 2025	Provisional patent application 2025906132 filed: explicit compositional reasoning in a continuous-present cognitive architecture.
19 Dec 2025	Provisional patent application 2025906680 filed: admissibility-controlled epistemic governance for machine reasoning.
27 Feb 2026	aurora-lens enters public PyPI release history (v0.1.1).
19 Mar 2026	aurora-lens v2.0.0 released on PyPI.
22 Mar 2026	Aurora-Lens live deployment producing runtime admissibility decisions with forensic audit output.
30 May 2026	PCT application PCT/AU2026/050569 filed: Admissibility-Controlled Epistemic Governance for Machine Reasoning Systems.
Jul 2026	The separate aurora-demo and Aurora-PEF histories were merged into the current release ancestry; GitHub web-flow signatures for the aurora-demo root chain were verified 15 Jul 2026 (Artefact 8).

Artefact 1 — Conceptual Bridges Archive

Archive repository containing early materials on machine understanding, conceptual bridges, and present-centred cognition. Repository state recorded 20 November 2025. Establishes conceptual development prior to the named architecture.

**milarien**

Private

Watch 0

main

1 Branch

0 Tags

Go to file

Add file

Code

**milarien**

Add files via upload

79b0a2d · 5 months ago

6 Commits

 Conceptual Bridges.docx	Add files via upload	5 months ago
 Conceptual Bridges.rtf	Add files via upload	5 months ago
 Philosophical Architecture of Machine Underst...	Add files via upload	5 months ago
 Philosophical Architecture of Machine Underst...	Add files via upload	5 months ago
 Present Centred Cognition Model.docx	Add files via upload	5 months ago
 README.md	Update README.md	5 months ago

 README

These are my own reflections on artificial intelligence — how language models process meaning, develop structure, and mirror aspects of human thought.

File: 01_conceptual_bridges_archive.png SHA-256: 014382a1554f623eae35f123ffba12d874e5b65826784fd0a2207c1c7db6c9ed

GitHub repository view: Conceptual Bridges Archive, 6 commits, containing Conceptual Bridges.docx/.rtf, Philosophical Architecture of Machine Understanding, Present Centred Cognition Model.docx, README.md.

Artefact 2 - Aurora-PEF Public README

Repository: Aurora-PEF. Root commit 05084259446ff42c3a524c74b6de24fc882b1330, created 4 December 2025 through GitHub's web interface. Its README names "Aurora + PEF - A Meaning-First Supervisory Architecture for Reliable AI Reasoning", identifies Margaret Stokes as author, and defines the Persistent Existence Frame, Roles, Domains, Spans, and the operators WE, THEN, WHILE, UNTIL, and BECAUSE.



milarien Create readme.md

9d3ca61 · 4 months ago

 1 Commit

 readme.md

Create readme.md

4 months ago

 README

Aurora + PEF

A Meaning-First Supervisory Architecture for Reliable AI Reasoning

Author: Margaret Stokes
Date: December 2025

Overview

Large language models generate fluent text, but they do not maintain conceptual structure. They collapse ambiguity, fabricate missing grammar or causality, and cannot revert once they have committed to an incorrect interpretation. These limitations are architectural, not behavioural, and they persist across GPT-4, Claude, Gemini, and Grok.

Aurora + PEF is a meaning-first supervisory reasoning architecture designed to integrate with existing LLMs. It provides the structural capabilities that transformers fundamentally lack while leaving expression to the underlying model.

Aurora introduces explicit conceptual entities — **Roles**, **Domains**, **Spans** — and deterministic compositional operators (WE, THEN, WHILE, UNTIL, BECAUSE).

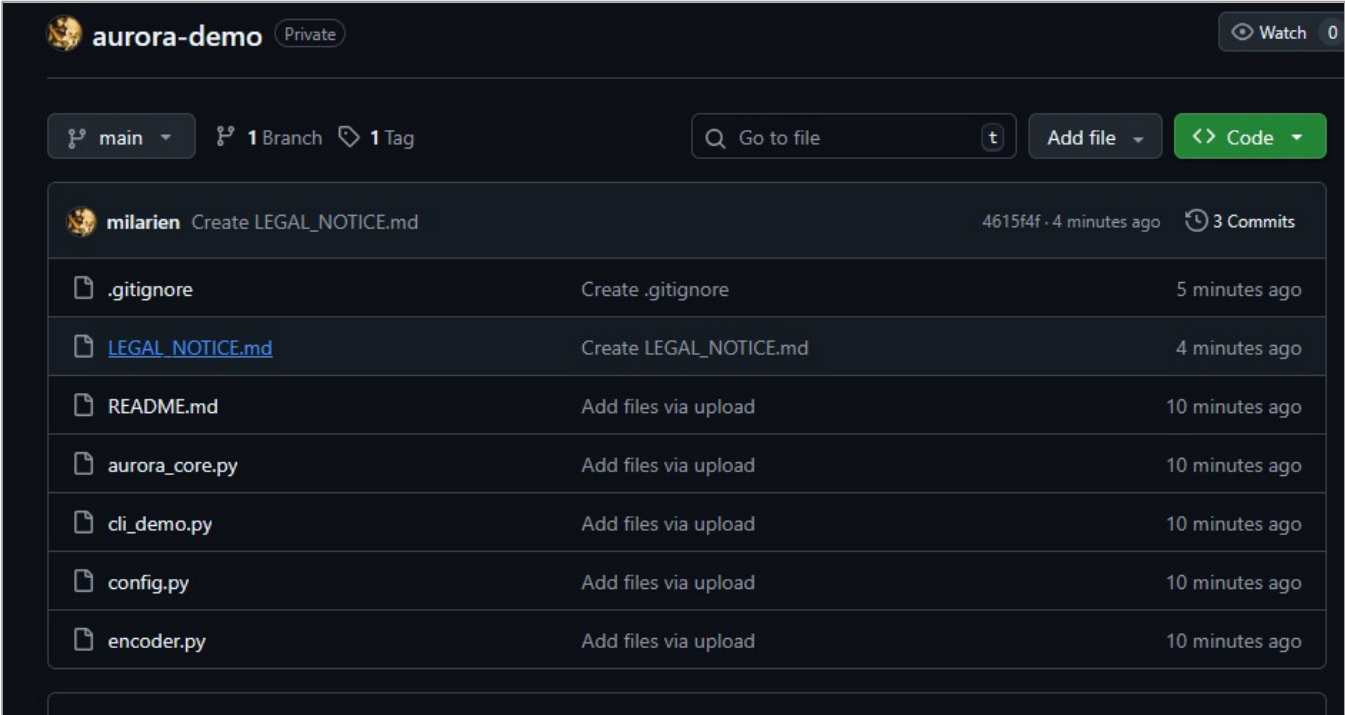
PEF, the **Persistent Existence Frame**, provides a non-temporal substrate that enables reversible inference and eliminates positional drift.

File: 02_aurora_pef_readme.png SHA-256: 48695a3c21e34dc9075dac874ffeb25575644e33708ff838059f1a499c63e674

Aurora-PEF README naming the architecture and author; root history begins 4 Dec 2025 at commit 0508425.

Artefact 3 — Aurora Demo Repository

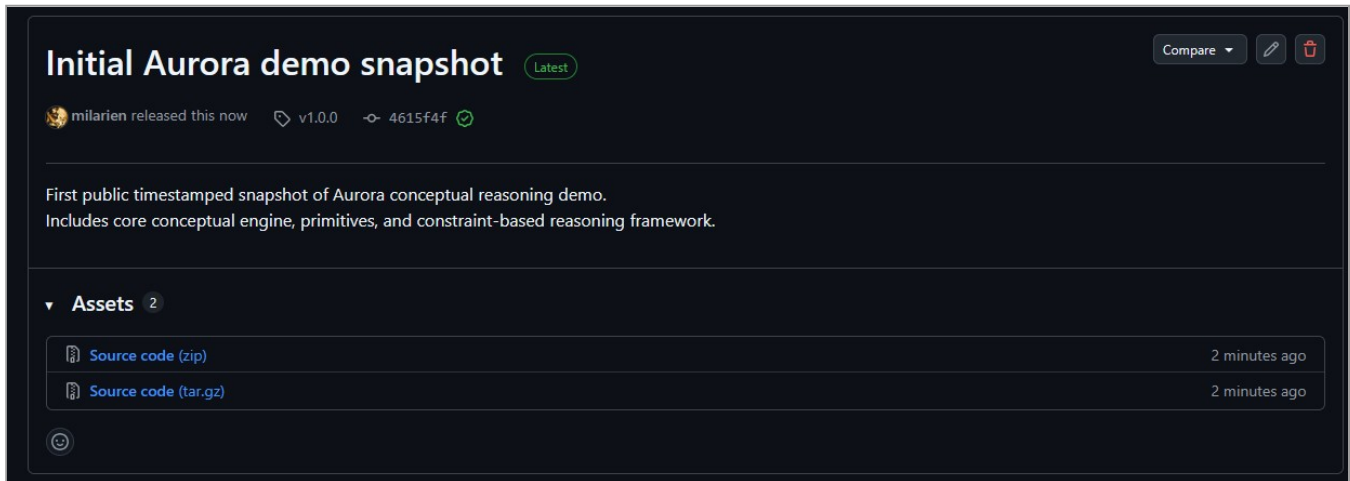
Working source repository (aurora-demo) showing implementation surface: `aurora_core.py`, `cli_demo.py`, `config.py`, `encoder.py`, `README.md`, `LEGAL_NOTICE.md`. Created 30 November 2025 via GitHub web interface; made public 6 December 2025.



File: 03_aurora_demo_repository.png **SHA-256:** eacff4afd3fbbef2b08c124185271db5fd79a3662430bf5018cb5b7281409398
Repository file listing minutes after creation; 3 commits, 1 tag.

Artefact 4 — Initial Public Release v1.0.0

Release "Initial Aurora demo snapshot", tag v1.0.0 on commit 4615f4f: "First public timestamped snapshot of Aurora conceptual reasoning demo. Includes core conceptual engine, primitives, and constraint-based reasoning framework."



File: 04_aurora_demo_release.png **SHA-256:** 8c5b4c0bdfe0baa6fd015ed0a33322279f6f36d60c2feb18c12087c1d44aa4e4
GitHub release page for v1.0.0 with source archives.

Artefact 5 — Direct Circulation Records

Sent-mail records showing direct circulation of the work to a bounded set of AI researchers, engineers, and public intellectuals at major laboratories. These records evidence transmission only — they do not imply endorsement, response, review, agreement, collaboration, or affiliation by any recipient.

☐	☆	To: Evan	A reasoning substrate for multi-hypothesis maintenance and constraint-driven collapse - Hi Evan, I've been running a set of controlled probes across Grok, GPT-4, Gemini, and Claude to t...	12/7/25	●
☐	☆	To: ryan	Reproducible structural ambiguity failure in LLMs + alternative reasoning substrate - Hi Ryan, Over the past month I've been running a small but clean empirical evaluation on a specific re...	12/6/25	●
☐	☆	To: jbt	A reproducible transformer limitation + the reasoning architecture that resolves it - Hi Josh, Over the past several weeks I've been running a series of controlled ambiguity tests across Gr...	12/6/25	✓
☐	☆	To: dieleman 2	Proof A reproducible transformer limitation + the reasoning architecture that resolves it - Hi Sander Over the past several weeks I've been running a series of controlled ambiguity tests a...	12/6/25	●
☐	☆	To: fchollet 2	Proof A reproducible transformer limitation + the reasoning architecture that resolves it - Hi Francois Over the past several weeks I've been running a series of controlled ambiguity tests ...	12/6/25	●
☐	☆	To: jared.kaplan	A reproducible transformer limitation + the reasoning architecture that resolves it - Hi Jared, Over the past several weeks I've been running a series of controlled ambiguity tests across G...	12/6/25	✓
☐	☆	To: gopnik	A reproducible transformer limitation + the reasoning architecture that resolves it - Hi Alison, Over the past several weeks I've been running a series of controlled ambiguity tests across G...	12/6/25	✓
☐	☆	To: slb 2	Proof A reproducible transformer limitation + the reasoning architecture that resolves it - Hi Sam, Over the past several weeks I've been running a series of controlled ambiguity tests acr...	12/6/25	✓
☐	☆	To: buck 2	Proof A reproducible transformer limitation + the reasoning architecture that resolves it - Hi Buck, Over the past several weeks I've been running a series of controlled ambiguity tests acr...	12/6/25	●
☐	☆	To: evan.hubing. 2	Proof A reproducible transformer limitation + the reasoning architecture that resolves it - Hi Evan Over the past several weeks I've been running a series of controlled ambiguity tests acr...	12/6/25	●
☐	☆	To: melanie 2	Proof A reproducible transformer limitation + the reasoning architecture that resolves it - Hi Melanie, Over the past several weeks I've been running a series of controlled ambiguity tests ...	12/6/25	●
☐	★	To: nelhage, nelson	A reproducible transformer limitation + the reasoning architecture that resolves it - Hi Nelson, Over the past several weeks I've been running a series of controlled ambiguity tests across ...	12/6/25	✓
☐	☆	To: melanie.mit. 2	Proof A reproducible transformer limitation + the reasoning architecture that resolves it - Hi Melanie, Over the past several weeks I've been running a series of controlled ambiguity tests ...	12/6/25	●
☐	☆	To: mm 2	Proof A reproducible transformer limitation + the reasoning architecture that resolves it - Hi Melanie, Over the past several weeks I've been running a series of controlled ambiguity tests ...	12/6/25	●
☐	★	To: paulfchrist.	A reproducible transformer limitation + the reasoning architecture that resolves it - Hi Paul, Over the past several weeks I've been running a series of controlled ambiguity tests across Gro...	12/6/25	✓
☐	☆	To: chris.olah	A reproducible transformer limitation + the reasoning architecture that resolves it - Hi Chris, Over the past several weeks I've been running a series of controlled ambiguity tests across Gr...	12/6/25	●
☐	☆	To: chris.olah 2	Proof A reproducible transformer limitation + the reasoning architecture that resolves it - Hi Chris, Over the past several weeks I've been running a series of controlled ambiguity tests acr...	12/6/25	●

File: 05_direct_circulation.png SHA-256: 3ca94877b59c6534ff3e9a86696c1f2e55a7b0ba1cb4d833fcfb810f3472595d

Sent-mail listing, 6–7 Dec 2025, subjects: "A reproducible transformer limitation + the reasoning architecture that resolves it" and variants.

Artefact 6 — Repository Clone Activity

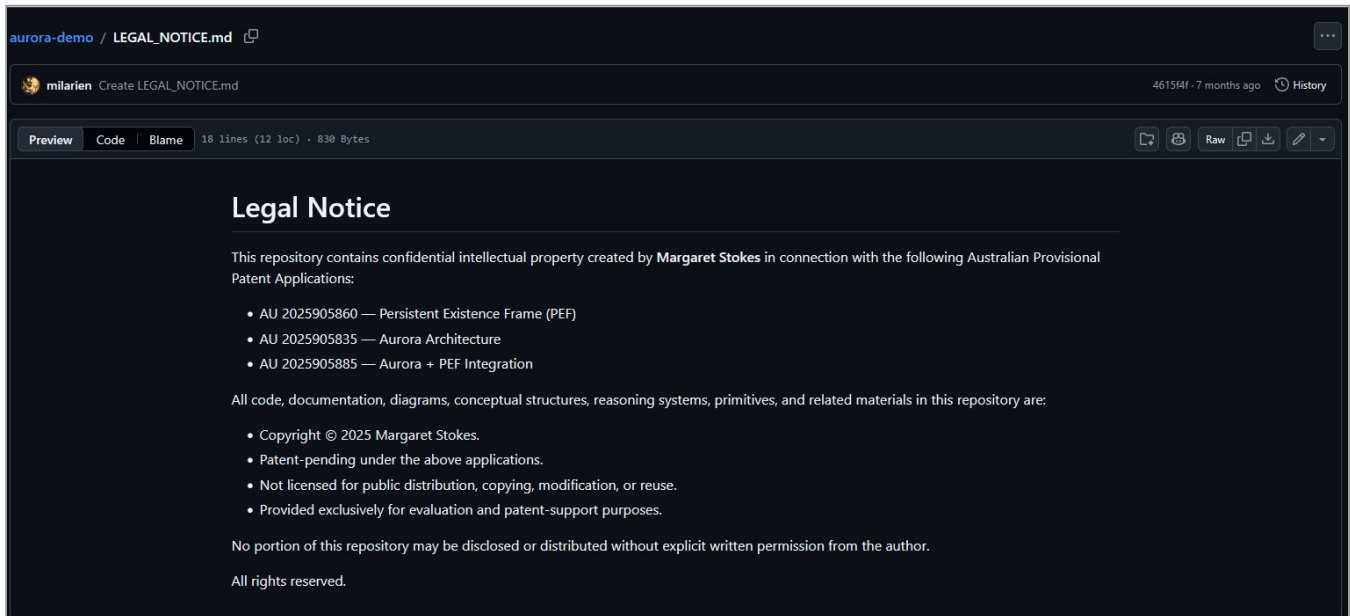
During the public exposure period, GitHub recorded **556 clone events from 324 unique cloning sources** in fourteen days, against a very small initial recipient group. Peak single-day activity: 166 clones (19 Dec), 97 unique cloners.



File: 06_repository_clone_activity.png SHA-256: ee0b4153e9c3810c4012454d7225378b13b06dabc26b6a8433a9cd0744faa440
GitHub traffic panel: "Clones in last 14 days — 556 Clones"; "Unique cloners in last 14 days — 324 Unique cloners".

Artefact 7 — Legal Notice

LEGAL_NOTICE.md created inside the repository on 30 November 2025, naming Australian Provisional Patent Applications AU 2025905860 (PEF), AU 2025905835 (Aurora Architecture), AU 2025905885 (Aurora + PEF Integration); asserting copyright © 2025 Margaret Stokes; restricting distribution prior to public release.



File: 07_legal_notice.png **SHA-256:** 7965db289e72e1abdce2c1741af411245c37a403c94ae8f6387c88b370eb26bb
Rendered LEGAL_NOTICE.md as committed 30 Nov 2025 (commit 4615f4f).

Artefact 8 — Platform-Verified Commits

The three root commits of 30 November 2025 were created through GitHub's web interface. GitHub is recorded as committer on each, each carries GitHub's **Verified** badge, and each commit object is cryptographically signed by GitHub's web-flow signing key. Verification performed 15 July 2026 against the key published at github.com/web-flow.gpg returned **Good signature** for all three.

Repository: aurora-demo (merged into aurora-lens-release master, July 2026)

Signing key: RSA B5690EEEBB952194 — GitHub <noreply@github.com>

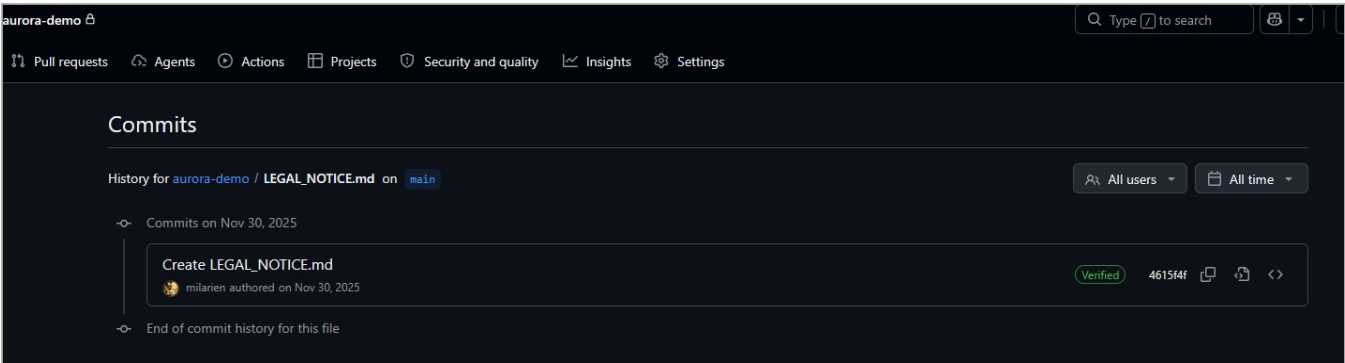
Key fingerprint: 9684 79A1 AFF9 27E3 7D1A 566B B569 0EEE BB95 2194

Commit 1: a35a461248cb53f41af49b00c1ccdd80e49f1175
30 Nov 2025 10:43:16 +1100 — "Add files via upload" — Good signature

Commit 2: 011f03b1180552f15894679f434488b4ef5c8256
30 Nov 2025 10:47:34 +1100 — "Create .gitignore" — Good signature

Commit 3: 4615f4f56e22f0be89fc941321fd78815a3377b0
30 Nov 2025 10:49:08 +1100 — "Create LEGAL_NOTICE.md" — tag v1.0.0 — Good signature

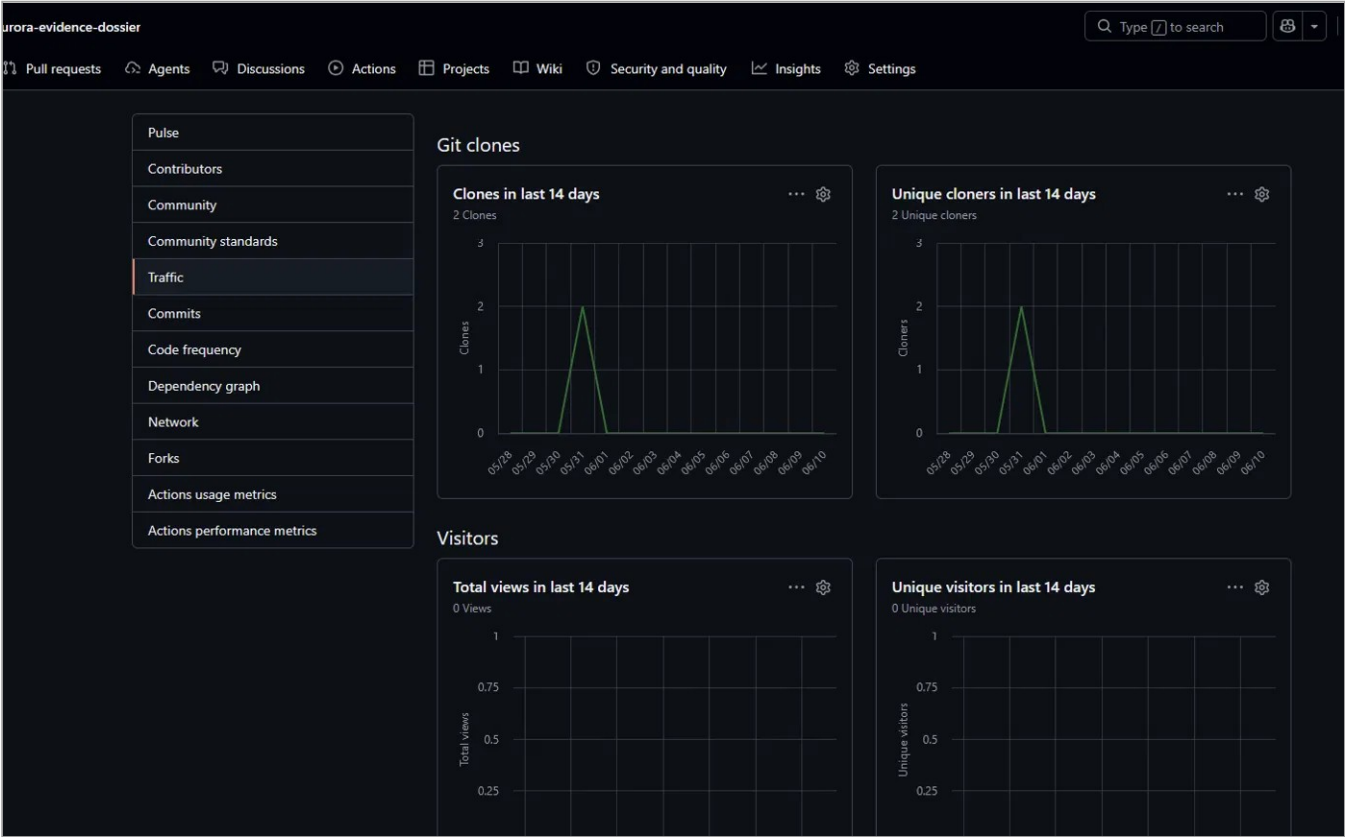
Because the signatures are embedded in the aurora-demo commit objects, the recorded dates, author fields, and content of those commits cannot be changed without invalidating the signatures and descendant hashes in that provenance chain. This proof applies to aurora-demo; Aurora-PEF is a separate repository history merged independently. Verification is repeatable from the repository with `git show --show-signature <hash>` after importing `github.com/web-flow.gpg`. Full transcript: `PROVENANCE_COMMIT_PROOF.txt`.



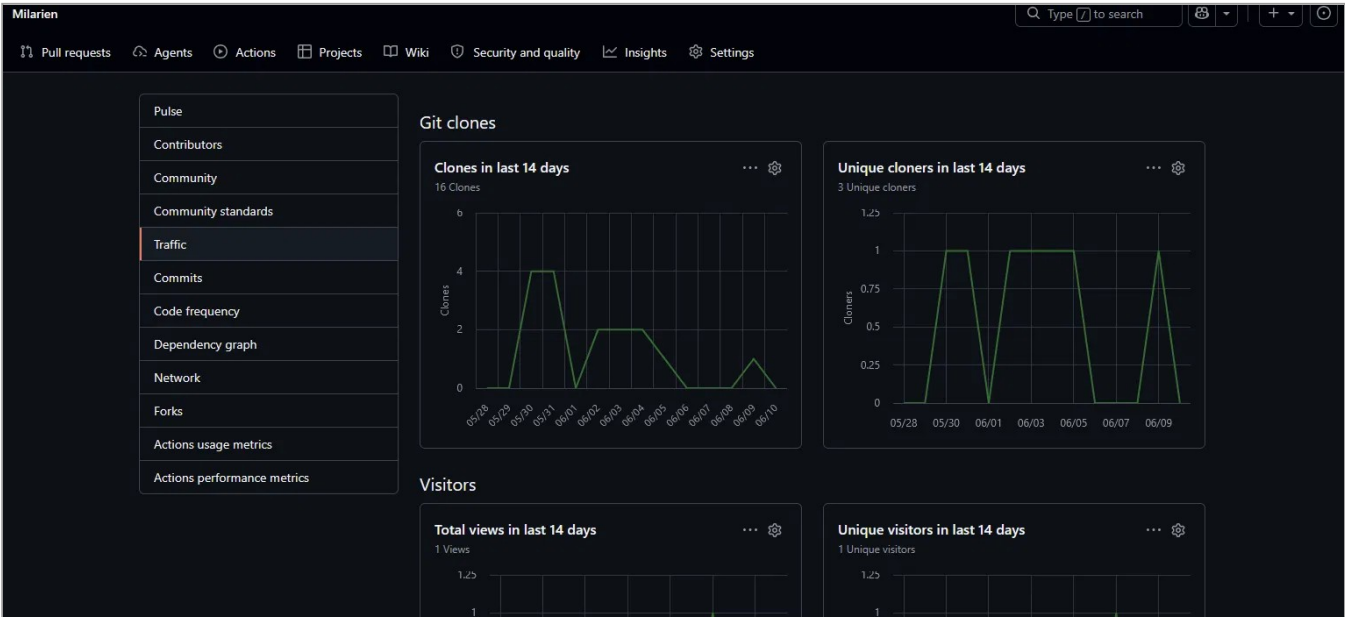
File: commit_08.png **SHA-256:** 5b12a1816230108defdbb3a46d4d730dbc0388dfc3d5fc04ae60de11625e39f4
GitHub commit history for LEGAL_NOTICE.md showing the Verified badge on commit 4615f4f, authored 30 Nov 2025.

Artefact 9 — Comparative Baselines

Control repositories on the same account over equivalent 14-day windows recorded 16 clones from 3 unique sources, and 2 clones from 2 unique sources respectively. These baselines establish normal traffic against which Artefact 6 is a two-orders-of-magnitude anomaly in unique sources.



File: comparative_repo_traffic.webp SHA-256: 206c64ca53bba6324d713f6c0a1129a8ea8ab09ffe2de5386265739834ead113
Baseline A: 16 clones / 3 unique cloners; 1 view / 1 unique visitor.



File: comparative_repo_02.webp SHA-256: c8f554e1516476ae4ca9ad0617b8fd6a60af8e638e8e226755cfb93200124c94
Baseline B: 2 clones / 2 unique cloners; 0 views.

Evidence manifest

SHA-256 digests of the artefact files as deposited in the supplementary evidence archive accompanying this dossier. Any party can confirm file integrity by recomputing the digest.

FILE	DATE	SHA-256	EVIDENTIARY ROLE
01_conceptual_bridges_archive.png	20 Nov 2025	014382a1554f623e ae35f123ffba12d8 74e5b65826784fd0 a2207c1c7db6c9ed	GitHub repository view of the Conceptual Bridges Archive; establishes prior conceptual development.
02_aurora_pef_readme.png	4 Dec 2025	48695a3c21e34dc9 075dac874ffeb255 75644e33708ff838 059f1a499c63e674	Aurora-PEF README naming the architecture and author; root commit 0508425 establishes the separate repository history.
03_aurora_demo_repository.png	30 Nov 2025	eacff4afd3fbbef2 b08c124185271db5 fd79a3662430bf50 18cb5b7281409398	
04_aurora_demo_release.png	30 Nov 2025	8c5b4c0bdfc0baa6 fd015ed0a3332227 9f6f36d60c2feb18 c12087c1d44aa4e4	v1.0.0 release "Initial Aurora demo snapshot" on commit 4615f4f; dated public release record.
05_direct_circulation.png	6-7 Dec 2025	3ca94877b59c6534 ff3e9a86696c1f2e 55a7b0ba1cb4d833 fcfb810f3472595d	Sent-mail records of direct circulation to named AI researchers and engineers; evidences targeted transmission to a bounded recipient set.
06_repository_clone_activity.png	6-19 Dec 2025	ee0b4153e9c3810c 4012454d7225378b 13b06dabc26b6a84 33a9cd0744faa440	GitHub traffic panel: 556 clones, 324 unique cloning sources in 14 days; the dissemination anomaly.
07_legal_notice.png	30 Nov 2025	7965db289e72e1ab dce2c1741af41124 5c37a403c94ae8f6 387c88b370eb26bb	LEGAL_NOTICE.md in aurora-demo naming provisional applications AU 2025905860, AU 2025905835, AU 2025905885 and asserting copyright.
commit_08.png	30 Nov 2025	5b12a1816230108d efdbb3a46d4d730d bc0388dfc3d5fc04 ae60de11625e39f4	GitHub commit history for LEGAL_NOTICE.md showing the Verified badge on commit 4615f4f, authored 30 Nov 2025.
comparative_repo_traffic.webp	control window	206c64ca53bba632 4d713f6c0a1129a8 ea8ab09ffe2de538 6265739834ead113	Comparative baseline: 16 clones / 3 unique cloners over an equivalent 14-day window on the same account.
comparative_repo_02.webp	control window	c8f554e1516476ae 4ca9ad0617b8fd6a 60af8e638e8e2267 55cfb93200124c94	Comparative baseline: 2 clones / 2 unique cloners over an equivalent 14-day window on the same account.
PROVENANCE_COMMIT_PROOF.txt	15 Jul 2026	527cc55bd49a0cf9 236c389345a692e9	Git ancestry and cryptographic signature verification record for the November 2025

		3206b757ac8fac78 8ef87c93ca3307dd	commits.
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Public research record

- ORCID: 0009-0004-6422-4174
- Zenodo — PCCM preprint: zenodo.org/records/18976303
- Zenodo — Epistemic governance: zenodo.org/records/18653120
- Zenodo — OECD alignment: zenodo.org/records/18719033
- OSF: osf.io/86bxj · SSRN author page: [per_id 10385176](https://ssrn.com/author/per_id/10385176)
- PyPI: pypi.org/project/aurora-lens

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